OXFORD
INTERNATIONAL
AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname ______________________________

Forename(s) ______________________________

Candidate signature ______________________________

INTERNATIONAL GCSE
BIOLOGY
Paper 2

Tuesday 29 May 2018   07:00 GMT   Time allowed: 1 hour 30 minutes

Materials
For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice
In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	

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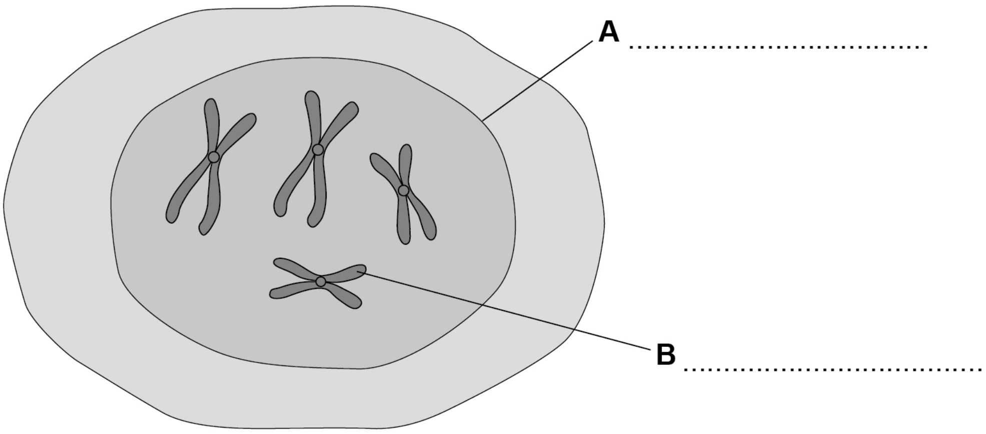
Answer **all** questions in the spaces provided.

0 1 **Figure 1** shows a body cell.

Figure 1



0 1 . 1 Label structures **A** and **B**. Use words from the box.

chromosome	**mitochondrion**	**nucleus**	**ribosome**	**vacuole**

[2 marks]

0 1 . 2 Egg cells and sperm cells are sex cells.

What is the scientific name for sex cells?

[1 mark]

Tick **one** box.

Clones ☐

Embryos ☐

Gametes ☐

Genotypes ☐

0 1 . 3 An egg cell and a sperm cell join to produce a single new cell.

What is this process called?

[1 mark]

Tick **one** box.

Differentiation ☐

Fertilisation ☐

Growth ☐

Speciation ☐

0 1 . 4 The new cell divides many times.

What is this process called?

[1 mark]

Tick **one** box.

Meiosis ☐

Mitosis ☐

Mutation ☐

Variation ☐

Question 1 continues on the next page

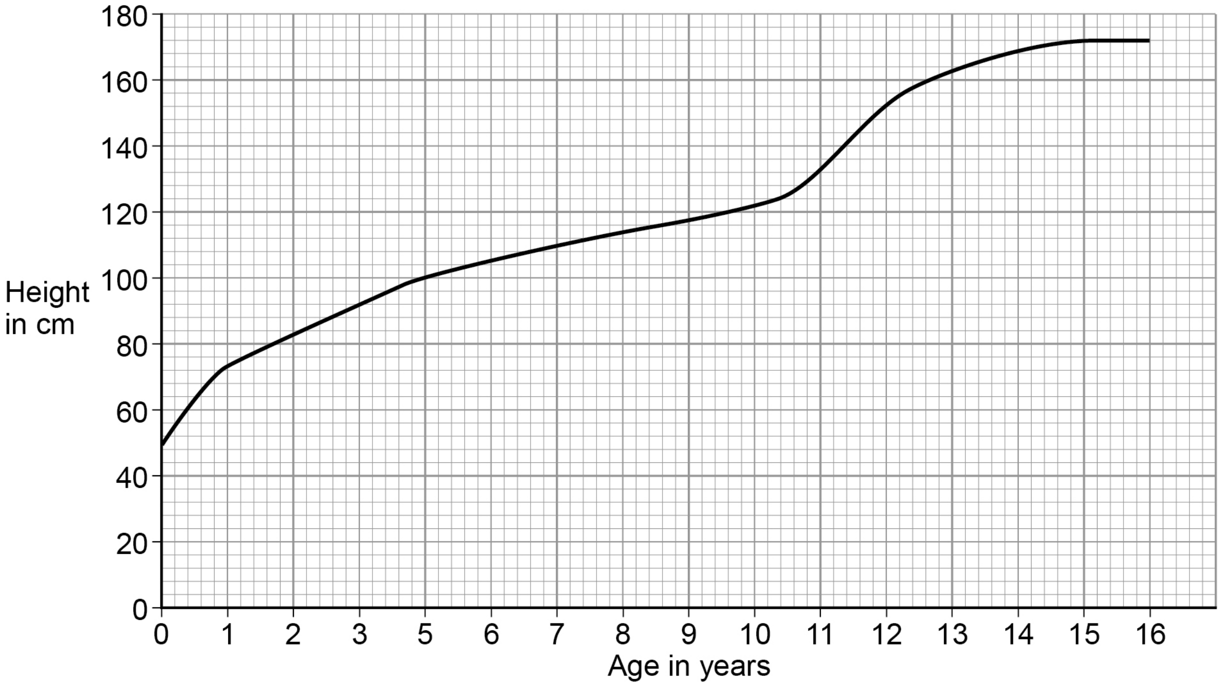
Turn over ►

Cell division continues as an organism grows.

Growth in humans can be measured as an increase in height.

Figure 2 shows the height of a girl from birth (0 years) to 16 years.

Figure 2



0 1 . 5 **Figure 2** shows the girl grew rapidly between birth and 1 year.

Then the girl grew steadily until a second phase of rapid growth.

At what age did this second phase of rapid growth start?

[1 mark]

_______________________________ years

Do not write outside the box

0 1 . 6 At what age did the girl stop growing in height?

[1 mark]

____________________________________ years

0 1 . 7 New cells are still produced even when the girl stops growing.

State **one** use for these new cells **other** than for growth.

[1 mark]

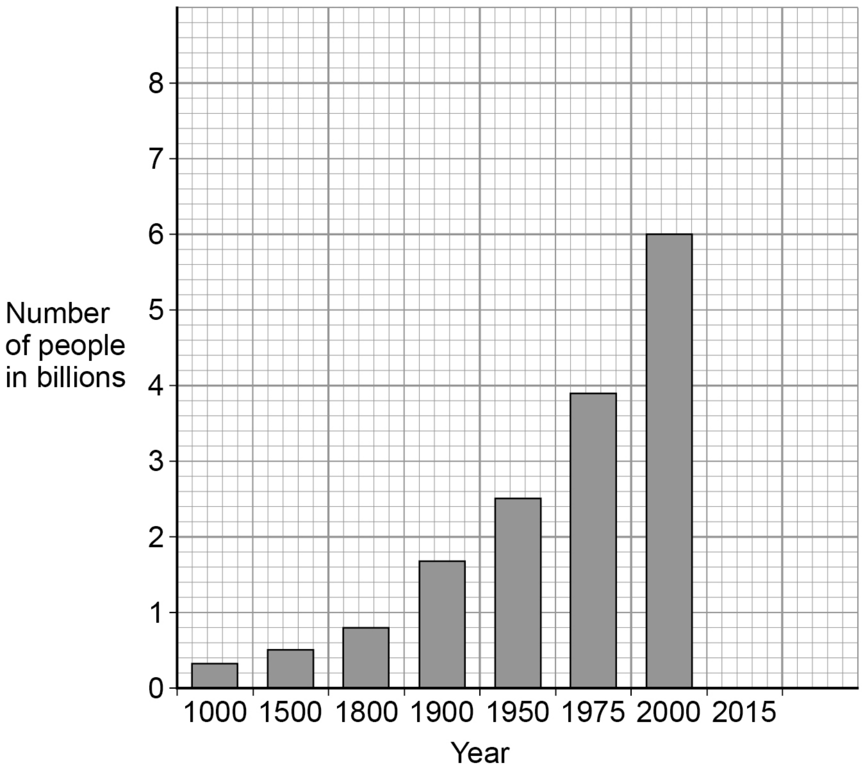
____

8

Turn over for the next question

0 2 **Figure 3** shows the increase in the human population from the years 1000 to 2015.

Figure 3



0 2 . 1 Calculate the increase in human population from 1800 to 2000

[2 marks]

Increase in population = ____________________ billion

0 2 . 2 In 2015 the human population was recorded as 7.2 billion.
Complete the bar chart in **Figure 3** for 2015.

[1 mark]

Do not write outside the box

0 2 . 3 Suggest **two** reasons why the human population increased rapidly from 1800 onwards.

[2 marks]

1 ____

____

2 ____

____

Deforestation is the removal of trees which affects global warming.
An increase in the human population has led to an increase in deforestation.

0 2 . 4 Which gas contributes to global warming?

[1 mark]

Tick **one** box.

Carbon dioxide ☐

Nitrogen ☐

Oxygen ☐

Sulfur dioxide ☐

0 2 . 5 What is global warming?

[1 mark]

____

____

Question 2 continues on the next page

Do not write outside the box

0 2 . 6 Describe how the removal of trees contributes to global warming.

[2 marks]

0 2 . 7 Deforestation increases the amount of dead plant matter in an ecosystem.
Explain how this will contribute to global warming.

[4 marks]

13

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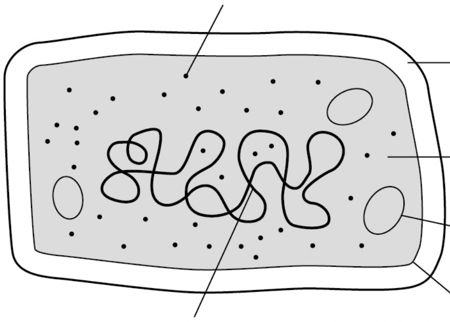
0 3 Bacterial cells and plant cells share many features.

Figure 4 shows a bacterial cell.

Figure 4



0 3 . 1 Name **two** structures that are present in **both** a bacterial cell and a plant cell.

[2 marks]

1 ____________________

2 ____________________

0 3 . 2 Bacteria can cause gum disease.

Gum disease can be treated with antibiotics.

How do antibiotics work?

[1 mark]

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0 3 . 3 A new antibiotic is used to treat gum disease.

Explain how a population of bacteria which are resistant to the antibiotic can develop.

[3 marks]

Do not write outside the box

Plant products such as blueberry juice and coconut oil can be used to treat gum disease.

A scientist investigated the antibacterial properties of these plant products.

This is the method used.

1. Heat a glass Petri dish containing agar at 120 °C for 30 minutes.
2. Allow the Petri dish to cool.
3. Spread bacteria over the agar surface in the Petri dish.
4. Soak three paper discs in either blueberry juice, coconut oil or antibiotic.
5. Place the paper discs on top of the agar.
6. Incubate the Petri dish at 37 °C for 48 hours.

0 3 . 4 The glass Petri dish and agar were heated at 120 °C to kill any bacteria present.
Why is it important to kill any bacteria present?

[1 mark]

____

____

0 3 . 5 Why was the Petri dish incubated at 37 °C?

[1 mark]

____

____

0 3 . 6 There are health risks when doing investigations with bacteria.

Give **two** ways risks to health can be reduced.

[2 marks]

1 ____

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2 ____

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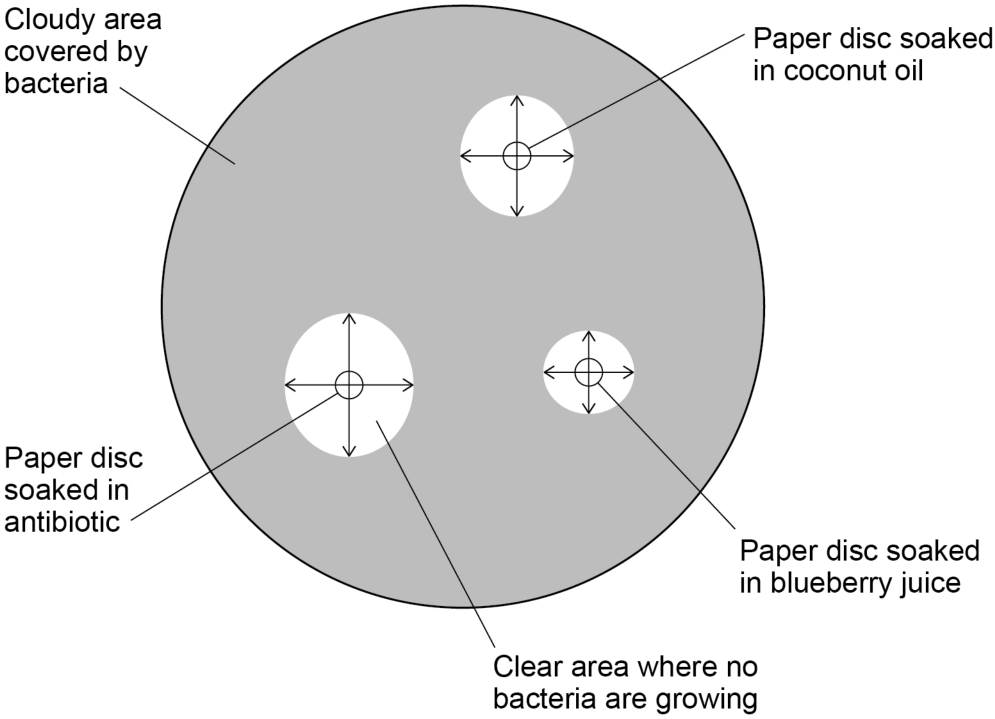
Turn over ►

Figure 5 shows what the Petri dish looked like after 48 hours.

Figure 5



0 3 . 7 **Table 1** shows some of the scientist's results.

Table 1

Test solution	Diameter of clear area in mm		Mean diameter of clear area in mm
	First reading	Second reading	
Antibiotic	17	19	18.0
Blueberry juice	12	11	11.5
Coconut oil			

Complete **Table 1** for the **coconut oil** using **Figure 5**.

- Measure the diameter of the clear area at **two** points.
- Calculate the mean diameter of the clear area.

[2 marks]

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0 3 . 8 The mean diameter of the clear area was used to assess antibacterial activity. The larger the diameter, the greater the antibacterial activity.

- Clear diameters of >13 mm had high antibacterial activity
- Clear diameters of 9 –13 mm had medium antibacterial activity
- Clear diameters of $<$ 9 mm had no antibacterial activity.

The scientist concluded that both blueberry juice and coconut oil could be used as effective antibacterial agents.

Do the results support the scientist's conclusion?

Give reasons for your answer.

Use data from **Table 1**.

[3 marks]

0 3 . 9 Another scientist repeated the same investigation a number of times.

The results were different from those shown in **Table 1**.

Suggest **two** reasons why.

[2 marks]

1

2

17

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0 4 . 1 Below are some examples of animal behaviour.

Draw **one** line from each example to the type of animal behaviour it demonstrates.

[3 marks]

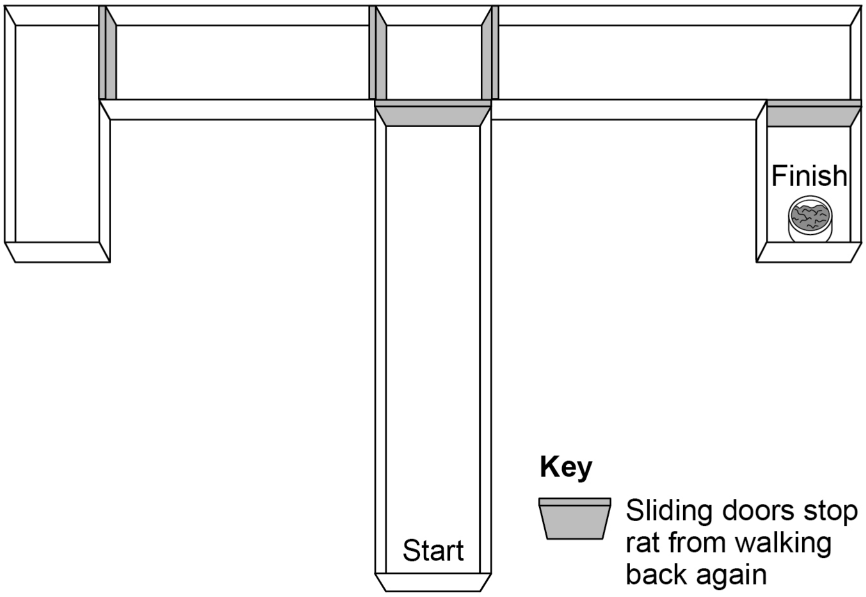
Example	**Type of animal behaviour**
A chick will follow the first moving object it sees as it hatches. After this it will only follow something that looks like the first object.	Classic conditioning
If a baby salamander is kept on its own away from water as it develops, when it is placed in water it will swim.	Habituation
Birds fly away when a farmer first fires a gun. If the farmer continues to fire the gun many times the birds will eventually ignore the noise.	Imprinting
	Innate
	Operant conditioning

Question 4 continues on the next page

0 4 . 2 A scientist investigated if giving more food to rats encouraged them to learn at a faster rate. The scientist used a T-maze.

Figure 6 shows a T-maze.

Figure 6



The scientist found that:

After 1 day

- 58% of the rats chose the correct route no matter how much food they received at the finish.

After 5 days

- 70% of the rats which received one piece of food at the finish chose the correct route.
- 90% of the rats which received four pieces of food at the finish chose the correct route.

Describe a method using the T-maze that the scientist could use to get these results.

[4 marks]

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0 4 . 3 Explain how the results demonstrate operant conditioning.

[3 marks]

Question 4 continues on the next page

Table 2 gives information about reproduction in three animals.

Table 2

	Stickleback	**Turtle**	**Lion**
Type of animal	Fish	Reptile	Mammal
Number of young per breeding cycle	40	150	3
Parental behaviour	Adult keeps eggs and young in mouth	Eggs are buried in the sand and left to hatch	Young develop inside the mother's womb until birth
Aftercare of offspring	1 week	None	2.5 years
Offspring survival rate	30%	5%	50%

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0 4 . 4 Explain how the number of young in each breeding cycle and parental care relates to the offspring survival rate.

You should include advantages and disadvantages for each animal.

[6 marks]

16

Turn over for the next question

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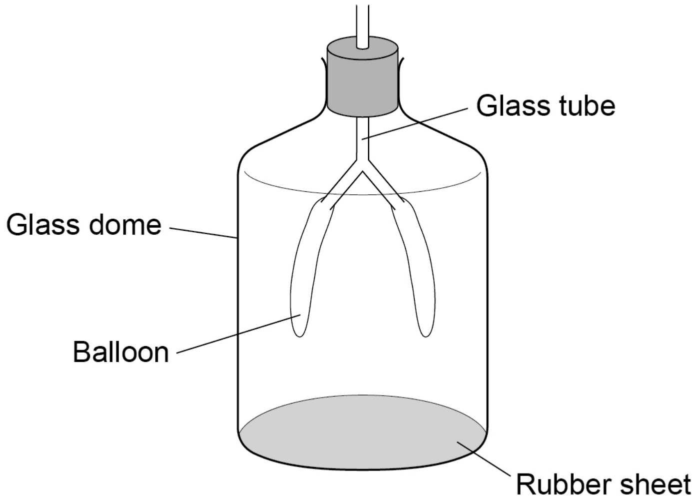
0 5 Breathing moves air into and out of the lungs.

Figure 7 is a model that can be used to show what happens when we breathe in and out.

Figure 7



0 5 . 1 Match the model parts labelled in **Figure 7** to the correct parts of the breathing system.

Draw **one** line from each model part to the correct part of the breathing system.

[4 marks]

Model part	**Part of breathing system**
	Alveolus
Balloon	Bronchiole
Glass dome	Diaphragm
Glass tube	Lung
Rubber sheet	Rib cage
	Trachea

Do not write outside the box

0 5 . 2 The balloons in **Figure 7** fill with air when the rubber sheet is pulled down.

Explain why air moves into the balloons.

[2 marks]

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0 5 . 3 Vital capacity measures the maximum volume of air that can be moved out of the lungs in one breath.

Calculate the vital capacity of a man with a height of 179 cm and mass of 70.0 kg

[3 marks]

First calculate the body surface area (BSA) using the following equation:

$$\text{BSA}\left(\text{m}^2\right) = \sqrt{\frac{\text{Height (cm)} \times \text{Mass (kg)}}{3600}}$$

Give your answer to three significant figures.

____

____

BSA = ____________________ m^2

The BSA can be used to calculate the vital capacity as follows:

$$\text{Vital capacity } (\text{cm}^3) = \text{BSA} \times 2500$$

Use this formula to calculate the vital capacity of the man.

____

____

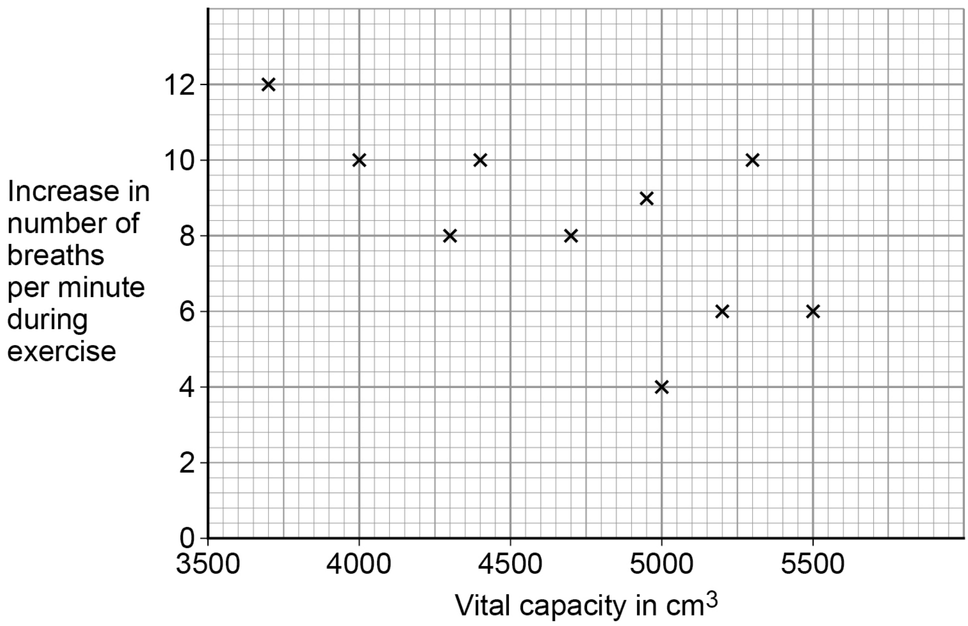
Vital capacity = ____________________ cm^3

Turn over ►

Ten men calculated their vital capacity. They measured their breathing rates (breaths per minute) before and during exercise.

Figure 8 shows their results.

Figure 8



0 5 . 4 Describe the relationship between vital capacity and increase in breathing rate during exercise.

Use data from **Figure 8**.

[3 marks]

Do not write outside the box

0 5 . 5 Explain the relationship you described in **05.4**.

[2 marks]

Question 5 continues on the next page

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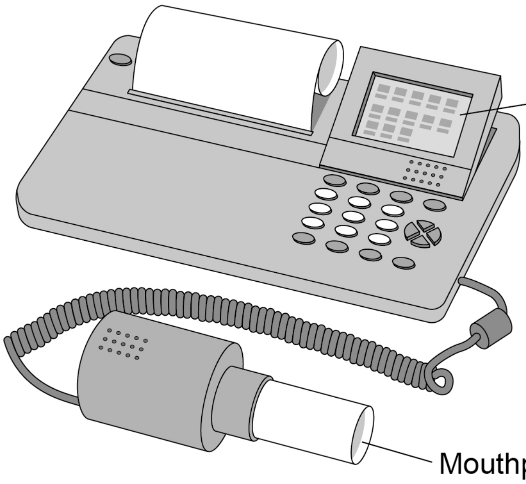
0 5 . 6 The men also used a spirometer to measure vital capacity.

Figure 9 shows a spirometer.

Figure 9



The men forced air out of their lungs and into the mouthpiece of the spirometer.
The spirometer recorded the amount of air that could be moved out of the lungs in one breath.

Figure 8 on page 22 shows a calculated vital capacity of 5300 cm^3 for one of the men.
The value recorded by this man on the spirometer was 4000 cm^3

Give **three** reasons why the spirometer reading was a more accurate estimate of vital capacity.

[3 marks]

1 ____

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2 ____

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3 ____

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17

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0 6 A scientist investigated how changes in carbon dioxide concentration and light intensity affected the growth of lettuce.

Table 3 shows the results.

Table 3

Carbon dioxide concentration in percentage	**Productivity of lettuce in arbitrary units**		
	10 000 Lux light intensity	**40 000 Lux light intensity**	**80 000 Lux light intensity**
0.02	12	21	27
0.04	14	24	33
0.06	15	29	42
0.08	1	30	46
0.10	15	30	47
0.12	12	28	46

0 6 . 1 The results in **Table 3** show how increasing carbon dioxide concentration and light intensity both affect the productivity of lettuce.

The maximum percentage (%) change in productivity compared with the value at 0.02% of carbon dioxide concentration can be calculated:

- at 10 000 lux of light the maximum change was +25%
- at 40 000 lux of light the maximum change was +43%

Calculate the maximum % change in productivity at 80 000 lux of light.

Give your answer to one decimal place.

[2 marks]

Percentage change = ____________________ %

Do not write outside the box

0 6 . 2 Productivity is a measure of growth.

Productivity is related to the rate of uptake of carbon dioxide. Explain why.

[3 marks]

0 6 . 3 A farmer used the results to determine the best conditions for growing lettuce to give maximum profit.

The farmer decided to use a light intensity of 80 000 lux and a carbon dioxide concentration of 0.06% in his greenhouses.

Evaluate the evidence for the farmer's decision.

[5 marks]

10

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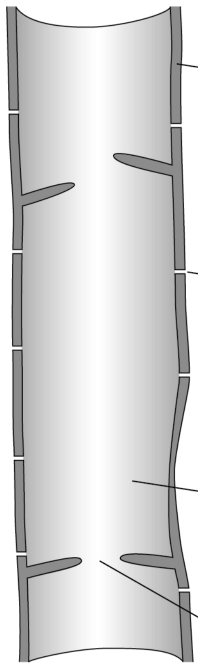
0 7 Xylem and phloem are plant tissues involved in transport.
Xylem is made of the remains of cells with no end walls that form tubes called xylem vessels.

Figure 10 shows a xylem vessel.

Figure 10



0 7 . 1 Suggest how the structure of a xylem vessel is related to its function.

[3 marks]

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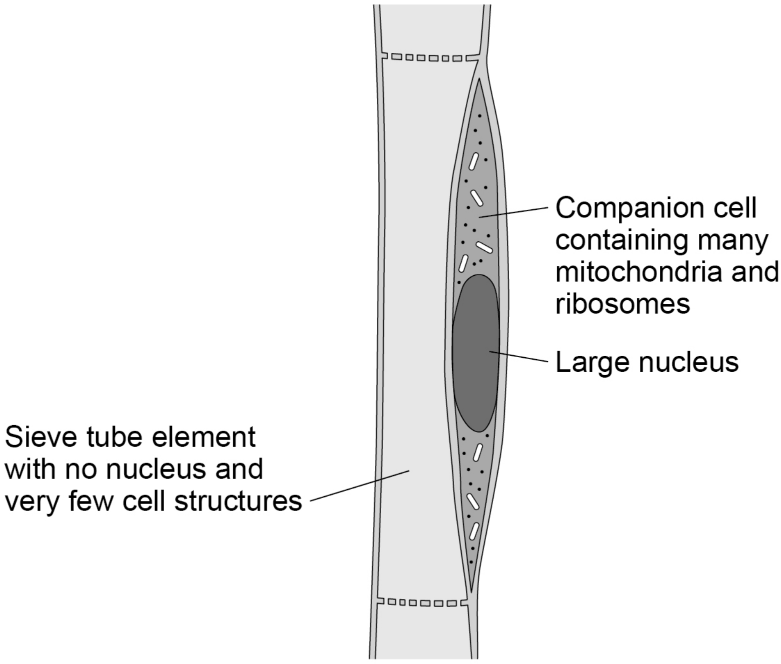
Phloem tissue consists of two types of cells:

- sieve tube elements
- companion cells.

Figure 11 shows phloem tissue.

Figure 11



0 7 . 2 There is one companion cell for every sieve tube element.
Suggest why the companion cell is essential for the successful functioning of the phloem tissue.

[3 marks]

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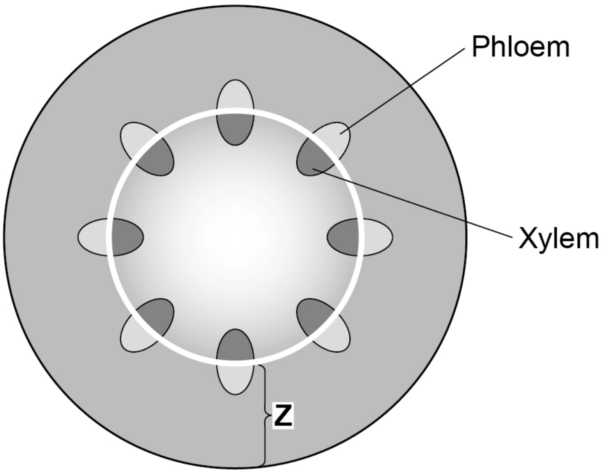
Turn over ►

Scientists investigated translocation through a plant stem.

Figure 12 shows a cross section of a plant stem.

Figure 12



This is the method used.

1. Label three similar plants, **A**, **B** and **C**.
2. Remove an outer ring of the tissues (labelled **Z** on **Figure 12**) from plants **B** and **C**.
3. Supply radioactive carbon dioxide to one leaf on plants **A** and **B**.
4. Leave plants **A** and **B** for 2 days so that sugars can form from the radioactive carbon dioxide.
5. Leave plant **C** for 1 month.

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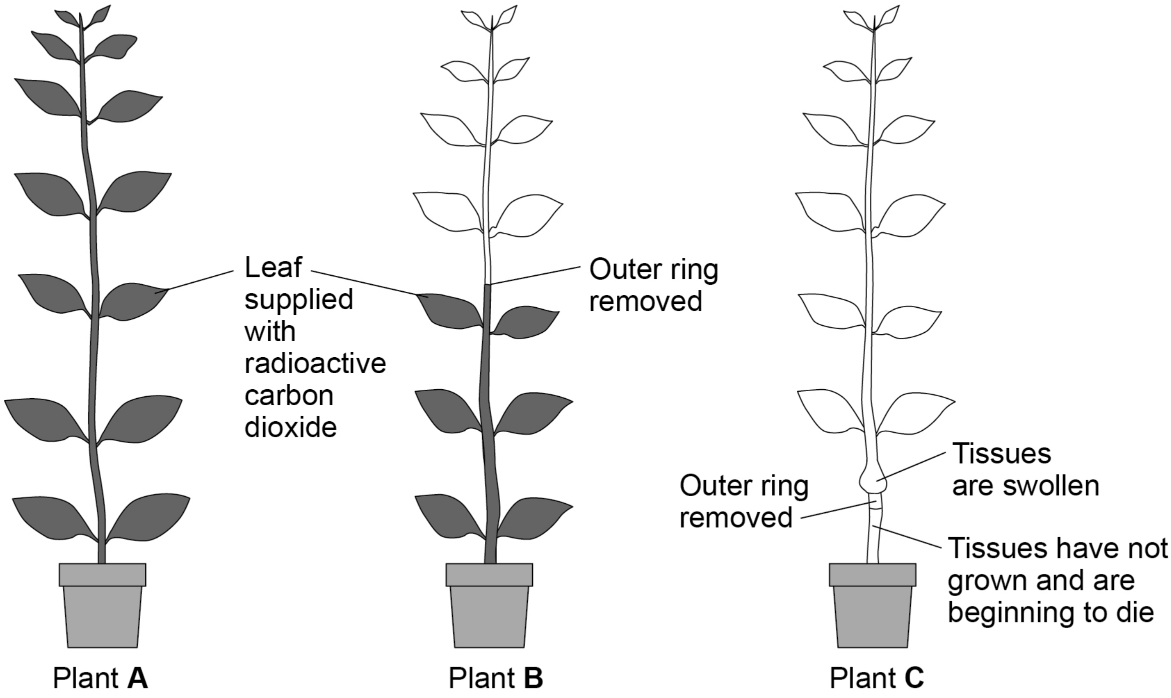
Figure 13 shows the results. The shaded areas in plants **A** and **B** show where radioactive sugar is detected.

Figure 13



0 7 . 3 Explain what the results for plants **A**, **B** and **C** show about translocation.

[3 marks]

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9

END OF QUESTIONS

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